Proposal: a learned stochastic developmental clock for STREAM

Abstract

Developmental stage is an imperfect proxy for the biological progress of an individual cell. A population collected at one stage can contain cells moving at different rates, yet precomputed pseudotime would introduce a heuristic coordinate outside the dynamics model. We propose instead to augment STREAM with a learned stochastic clock. A UCE-conditioned scalar process controls how far a cell travels along a normalized, gene-valued regulatory velocity direction, while a separate residual diffusion accounts for variation not explained by progress rate. The clock is learned from unpaired endpoint populations by differentiable rollout and requires neither pseudotime labels nor absolute-time conditioning. This document distinguishes the clock from the current gene-wise score SDE, defines identifiable parameterizations and positive increments, describes fitting and validation, and specifies ablations that test whether variable developmental progress is responsible for the current persistence-level forecasts.

1 Motivation

The causal STREAM model defines an autonomous expression-space field

$$\frac{\mathrm{d}x}{\mathrm{d}t} = v_\theta\big(E(x), \{Z_g\}_{g=1}^{G}\big)\,, \tag{1}$$

where $x \in \mathbb{R}^G_{\geq 0}$ is gene expression, E is the frozen UCE encoder, and Z_g contains AlphaGenome-derived regulatory tokens linked to gene g. The output remains one velocity per gene. Absolute developmental time is not an input; UCE is recomputed from the current expression state during rollout.

Equation 1 nevertheless assumes that the experimental time increment corresponds to the same amount of biological progress for every cell. That assumption is unlikely to hold exactly. Cells sampled at one stage may be delayed, rapidly differentiating, cycling, quiescent, or already committed to different fates. Diffusion pseudotime, Palantir, and related methods infer useful orderings from expression geometry [1, 2], while dynamical RNA-velocity models can infer gene-specific latent times from splicing kinetics [3]. However, supplying one of these coordinates to STREAM would make the model depend on a separately chosen graph, root, feature space, and smoothing procedure.

The desired alternative is to make progress itself latent and learn it jointly with the expression dynamics. Experimental time then specifies the duration over which a cell may progress, rather than directly fixing the distance it must travel through expression space.

2 Why the current SDE is not already a clock

The coupled score-flow model currently supports an SDE of the form

$$\mathrm{d}X_t = [v_\theta(X_t) + \kappa\Sigma s_\theta(X_t)]\,\mathrm{d}t + \sqrt{2\kappa}\,\Sigma\mathrm{d}W_t, \tag{2}$$

where Σ contains fixed gene scales. This noise can cause some particles to move farther than others, so it may incidentally create variation in apparent progress. It does not, however, encode progress explicitly:

- the Brownian increment has one component per gene and need not align with the developmental velocity;
- much of its variation can be perpendicular to the learned trajectory;
- the score correction is designed to control population marginals, not to estimate a cell-specific developmental rate; and
- a wider endpoint distribution can improve Sinkhorn distance without producing a coherent or interpretable ordering.

A developmental clock should instead perturb all genes coherently along the same local velocity direction. Residual gene-wise diffusion may still be useful, but it should be represented and evaluated separately.

3 Model

3.1 Latent stochastic clock

Introduce a scalar biological-progress process S_t and write

$$\mathrm{d}S_t = a_\phi(X_t)\,\mathrm{d}t + \sigma_\phi(X_t)\,\mathrm{d}W_t^S, \tag{3}$$

$$\mathrm{d}X_t = u_\theta(X_t)\,\mathrm{d}S_t + G_\psi(X_t)\,\mathrm{d}W_t^\perp. \tag{4}$$

The scalar clock rate and variability are state-dependent,

$$a_\phi(x) = \mathrm{softplus}(h_a(E(x)))\,, \qquad \sigma_\phi(x) = \mathrm{softplus}(h_\sigma(E(x)))\,, \tag{5}$$

where h_a and h_σ are small heads on frozen UCE state. The direction u_θ retains the gene-specific CRE-conditioned architecture. The same scalar realization $\mathrm{d}S_t$ multiplies every gene for one cell, so clock variation moves that cell coherently along its local developmental direction. $G_\psi \mathrm{d}W_t^\perp$ represents residual stochasticity such as unresolved fate choice or technical variation.

The clock is a latent dynamical variable, not a conditioning input. In particular, neither S_t nor absolute developmental time is supplied to u_θ, a_ϕ, or σ_ϕ. Every field is a function of the current expression-derived state.

3.2 Positive progress increments

A Gaussian Euler increment for Equation 3 can occasionally be negative. Small backward fluctuations may be biologically acceptable, but a first implementation should test a monotone clock using positive lognormal increments:

$$\Delta S_k = a_\phi(X_k)\Delta t \exp\left[\sigma_\phi(X_k)\sqrt{\Delta t}\,\epsilon_k - \frac{1}{2}\sigma_\phi(X_k)^2\Delta t\right], \qquad \epsilon_k \sim \mathcal{N}(0,1). \tag{6}$$

The correction in the exponent preserves $\mathbb{E}[\Delta S_k \mid X_k] = a_\phi(X_k)\Delta t$. The expression update is

$$X_{k+1} = \Pi_+\left[X_k + u_\theta(X_k)\Delta S_k + G_\psi(X_k)\sqrt{\Delta t}\,\epsilon_k^\perp\right], \tag{7}$$

where Π_+ projects expression to the nonnegative orthant. This reparameterization is differentiable with respect to the model parameters.

Gamma increments or an integrated positive stochastic rate can be considered later. The lognormal increment is simple, has a direct coefficient-of-variation interpretation, and requires one additional Gaussian draw per cell and step.

3.3 Direction normalization

Without a constraint, multiplying u_θ by a constant and dividing a_ϕ by the same constant leaves the expression dynamics unchanged. To separate direction from progress, normalize the raw regulatory velocity by a robust gene-space norm:

$$u_\theta(x) = \frac{v_\theta(x)}{\sqrt{G^{-1}\sum_{g=1}^{G}\left[v_{\theta,g}(x)/q_g\right]^2 + \epsilon}}, \tag{8}$$

where q_g is a fixed training-only robust gene scale. The clock increment then determines distance traveled in scaled expression space. A less invasive first experiment can freeze the pretrained v_θ and learn only the two clock heads; Equation 8 is preferable when the direction and clock are later trained jointly.

3.4 Separating clock and residual noise

Clock noise is rank one locally: its expression-space direction is $u_\theta(x)$. Residual diffusion should not be allowed to relearn the same component. Given an unconstrained residual draw r, project it away from the scaled velocity direction,

$$r_\perp = r - u\frac{\langle r, u\rangle_q}{\langle u, u\rangle_q}, \qquad \langle y, z\rangle_q = G^{-1}\sum_g \frac{y_g z_g}{q_g^2}. \tag{9}$$

This decomposition makes the interpretation testable: σ_ϕ controls variation in progress, while G_ψ controls variation transverse to the trajectory. The first ablation should omit G_ψ entirely so that any gain can be attributed to the clock.

4 Learning without pseudotime labels

4.1 Unpaired population objective

For each observed interval (t_0, t_1), independently sample source cells $\{x_i^0\}_{i=1}^B$ and target cells $\{x_j^1\}_{j=1}^C$. Roll M particles from each source according to Equation 7. No cell-level OT coupling is constructed. Compare the generated endpoint population $\widehat{\mathcal{X}}_1$ with the observed target population $\mathcal{X}_1$ using

$$\mathcal{L}_{\text{endpoint}} = S_\varepsilon\Big(P(\widehat{\mathcal{X}}_1), P(\mathcal{X}_1)\Big) + \lambda_m\mathcal{L}_{\text{mean}} + \lambda_C\mathcal{L}_{\text{cov}}, \tag{10}$$

where P is a differentiable, train-only log-normalized PCA map and S_ε is debiased Sinkhorn divergence. Gene means are compared after robust scale normalization, and covariance is compared in PCA space. This is the same population-level supervision used by the explicit score-flow fine-tuning implementation.

The experimental interval length enters only through the number and size of integration steps. The model learns how much biological progress usually occurs during that duration from endpoint distributions. It never receives a precomputed pseudotime value for an individual cell.

4.2 Clock-aligned flow and score matching

The clock can also be pretrained without differentiating through full rollouts. Standard flow matching by itself learns only the experimental-time expression field

$$b(x) = a_\phi(x)u_\theta(x). \tag{11}$$

It cannot identify the rate and direction separately: multiplying u_θ by a constant and dividing a_ϕ by the same constant leaves b unchanged. Direction normalization in Equation 8, a rate anchor, and supervision across multiple interval lengths are therefore required before interpreting a_ϕ as latent progress rate.

Given coupled endpoints, define the scaled chord length and direction

$$d = x_1 - x_0, \qquad \ell = \sqrt{G^{-1}\sum_g (d_g/q_g)^2}, \qquad \widehat{d} = d/(\ell + \epsilon). \tag{12}$$

A clock-aligned stochastic interpolant perturbs the linear path using one scalar Gaussian variable shared by every gene:

$$X_\tau = (1-\tau)x_0 + \tau x_1 + \gamma(\tau)\widehat{d}\, z, \qquad z \sim \mathcal{N}(0,1), \tag{13}$$

$$\partial_\tau X_\tau = d + \gamma'(\tau)\widehat{d}\, z. \tag{14}$$

Unlike the current gene-wise Gaussian interpolant, the random term changes distance along the endpoint chord rather than independently perturbing every gene. The model predicts a normalized direction $u_\theta(X_\tau)$, positive rate $a_\phi(X_\tau)$, clock scale $\sigma_\phi(X_\tau)$, and scalar noise $\widehat{z}_\psi(X_\tau, \tau)$. A coupled clock-flow readout has the form

$$f_{\text{clock}} = a_\phi u_\theta + \frac{\gamma'(\tau)}{t_1 - t_0}\sigma_\phi u_\theta \widehat{z}_\psi, \tag{15}$$

while the score component tangent to the trajectory is constructed from the same scalar estimate,

$$s_\parallel = -\frac{\widehat{z}_\psi}{\gamma(\tau)\sigma_\phi + \epsilon}\, u_\theta. \tag{16}$$

Thus the flow and score cannot disagree about the latent clock fluctuation. Because rank-one clock noise is singular in the full gene space, Equation 16 is a tangent score rather than a complete ambient score. A small fixed residual variance or the orthogonal diffusion in Equation 9 is needed if a nonsingular full-space score is required.

An initial simulation-free objective can combine

$$\mathcal{L}_{\text{clock-FM}} = \mathcal{L}_{\text{flow}} + \lambda_z \|\widehat{z}_\psi - z\|^2 + \lambda_d \left[1 - \cos_q(u_\theta, \widehat{d})\right] + \lambda_a (\log a_\phi)^2. \tag{17}$$

The direction term should be downweighted or omitted once the model is fine-tuned against population endpoints, because an individual endpoint chord is a coupling-dependent training device rather than a known lineage.

Simulation-free stochastic-interpolant training still requires an endpoint coupling. A large-pool partial OT or KL-relaxed UOT plan can supply diverse training paths without interpreting pairs as true descendants. Avoiding a fixed coupling entirely requires either the explicit unpaired rollout loss in Equation 10 or an iterative Schrödinger-bridge procedure that alternates SDE simulation and score fitting. The latter remains a flow/score method, but is more expensive and introduces an iterative bridge estimation loop.

Gaussian clock score matching corresponds naturally to $\mathrm{d}S = a\,\mathrm{d}t + \sigma\,\mathrm{d}W$ and therefore permits occasional backward increments. The strictly positive lognormal increment in Equation 6 does not have the same simple Gaussian denoising target. Consequently, the recommended hybrid is:

1. pretrain normalized direction, rate, and scalar clock score with clock-aligned stochastic-interpolant matching;
2. freeze or strongly anchor the normalized direction; and
3. fine-tune the positive stochastic clock with the unpaired endpoint rollout objective.

This uses flow/score matching for efficient local field estimation and the population objective for monotonicity and coupling-independent model selection.

4.3 Anchors and regularization

The endpoint objective alone admits weakly identified solutions. The initial loss should therefore include

$$\mathcal{L} = \mathcal{L}_{\text{endpoint}} + \lambda_a \mathbb{E}[(\log a_\phi)^2] + \lambda_\sigma \mathbb{E}[\sigma_\phi^2] + \lambda_\theta \|\theta - \theta_0\|_2^2. \tag{18}$$

The rate prior centers a_ϕ near one in the chosen progress units, the noise penalty prevents variance inflation from substituting for dynamics, and θ_0 is the simulation-free pretrained velocity checkpoint. These are soft anchors rather than pseudotime labels.

Validation must use cells disjoint from training within every observed stage, fixed source and target populations, fixed clock-noise draws, and an interval-balanced average. The pretrained fixed-clock model is validation step zero. Checkpoint selection should require a material improvement over that baseline rather than merely a decreasing training objective.

4.4 Truncated gradients through UCE

At each rollout step, UCE is recomputed from current raw expression using the low-variance systematic tokenization scheme. Tokenization is discrete and the encoder is frozen, so its state should be treated as stop-gradient. Gradients still reach every clock and velocity evaluation directly and pass through the additive expression updates. KNN or metacell smoothing may be used to construct simulation-free pretraining targets, but it should not be applied during causal clock rollout because a same-stage reference population is not available for a future cell.

5 Interpretation and limitations

5.1 What the clock can explain

The model is appropriate when cells share a local developmental path but vary in how quickly they move along it. It yields interpretable quantities:

- $a_\phi(x)$: expected progress rate for a cell state;
- $\sigma_\phi(x)$: uncertainty or heterogeneity in that rate;
- $S_{t_1} - S_{t_0}$: realized latent progress over an interval; and
- $u_\theta(x)$: gene-expression direction per unit progress.

Stage-wise distributions of inferred progress should be ordered on average, but overlap is expected and is one motivation for the model.

5.2 Branching

A scalar clock does not resolve branching when the same expression state can enter qualitatively different futures. Clock noise changes distance along a direction, not the direction itself. If the clock improves short-range forecasting but fails near fate decisions, the next extension should use a small mixture of regulatory velocity directions with a state-dependent gate, or retain an explicitly low-rank transverse diffusion. Branch-specific complexity should not be added until the rank-one clock ablation establishes that variable progress is useful.

5.3 Proliferation and changing population mass

Neither the clock nor a mass-preserving Sinkhorn loss explains changes in cell type abundance caused by division, death, or sampling. A progression-rate head may correlate with proliferation, but it does not create or remove particles. Growth weights, a learned birth–death rate, or unbalanced endpoint loss remain complementary. This distinction is important: faster progress, trajectory branching, and population growth are separate mechanisms and should not be absorbed into one noise parameter.

6 Ablation plan

Arm	Dynamics	Learned additions	Question
A	$\mathrm{d}X = v_\theta \mathrm{d}t$	None	Fixed-clock persistence-relative baseline
B	$\mathrm{d}X = a_\phi u_\theta \mathrm{d}t$	Deterministic rate	Does state-dependent speed help?
C	$\mathrm{d}X = u_\theta \mathrm{d}S$	Rate and scalar clock noise	Does variable progress improve endpoint prediction?
D	Current score SDE	Gene-wise score/noise	Is generic diffusion sufficient?
E	Clock plus $G_\psi \mathrm{d}W^\perp$	Clock and residual noise	Is transverse stochasticity additionally required?

Table 1: Staged ablations separating variable developmental progress from generic expression diffusion.

Arms A–C should be implemented first, using identical independent endpoint batches and Brownian seeds. Begin by freezing the pretrained regulatory velocity, then repeat B and C with normalized direction fine-tuning if either shows held-out benefit. Arm E is justified only if C improves progress-related metrics but leaves systematic multimodal endpoint errors.

7 Evaluation

The primary endpoint metric remains persistence-normalized Sinkhorn skill,

$$\text{skill} = 1 - \frac{S_\varepsilon(P(\widehat{\mathcal{X}}_1), P(\mathcal{X}_1))}{S_\varepsilon(P(\mathcal{X}_0), P(\mathcal{X}_1))}. \tag{19}$$

Positive skill means the predicted population is closer to the observed endpoint than a no-change forecast. This must be accompanied by gene-level mean-shift R^2, mean-shift MAE, per-gene Wasserstein distance, and major-cell- type-stratified metrics. Sinkhorn improvement alone can reward variance broadening.

Clock-specific diagnostics should include:

- distributions of a_ϕ, σ_ϕ, and realized ΔS by experimental stage and major cell type;
- the fraction of endpoint variance explained by clock-aligned versus transverse displacement;
- stability of rates under Brownian seed, particle count, and rollout step count;
- calibration of predicted progress variance against observed spread along the local velocity direction; and
- correlations with cell-cycle scores as diagnostics, not training labels, to detect a clock that merely recovers proliferation state.

Success requires a held-out-stage gain over the fixed-clock model, not only an observed-interval validation gain. The learned clock should improve both distributional and gene-shift metrics, remain stable across integration settings, and produce bounded state-dependent rates rather than extreme variance in a small number of cells.

8 Implementation plan

The implementation should preserve the existing separation between reusable model code and atlas-specific scripts:

1. Add reusable scalar rate and clock-noise heads operating on frozen UCE state. Initialize the deterministic rate near one and clock noise near zero.
2. Add the positive stochastic increment in Equation 6 to the differentiable population rollout module. Accept fixed scalar Gaussian draws for reproducible validation.
3. Implement robust direction normalization and verify invariance to a global rescaling of raw velocity.
4. Add the scalar clock-aligned stochastic interpolant and coupled flow/noise loss in Equations 13 and 17. Pretrain with matched large-pool partial-OT or UOT paths.
5. Fit Arm B and Arm C from the same pretrained score-flow or autonomous checkpoint, initially freezing the regulatory direction, then fine-tune with unpaired positive-clock rollouts.
6. Extend checkpoints with clock configuration, initialization, normalization scales, parent checkpoint, and selected validation step.
7. Extend held-out causal evaluation with fixed-clock, deterministic-rate, and stochastic-clock controls using identical source cells.
8. Add synthetic tests for known variable-speed trajectories, including a case where diagonal diffusion broadens the population but cannot match coherent progress variation.
9. Run matched raw, KNN-pretrained, and metacell-pretrained models without applying smoothing during rollout.

The first experiment should be small: two particles per source, two to four rollout steps, a deterministic-rate arm, and a stochastic-clock arm. The existing explicit population fine-tuning workflow already supplies independent endpoint batches, differentiable PCA/Sinkhorn loss, frozen validation draws, and pretrained checkpoint anchoring. The main new machinery is therefore two scalar heads and one shared random increment per cell and step.

9 Decision rule

The stochastic clock is supported if Arm C improves held-out endpoint skill and gene-shift metrics over both Arm A and Arm B, and if learned progress variance is nonzero but bounded. If Arm B improves while Arm C does not, cells benefit from state-dependent deterministic rates but stochastic pseudotime is unnecessary. If the current gene-wise SDE improves while the clock does not, the useful uncertainty is not primarily variation in developmental progress. If no arm exceeds persistence, the limiting issue is more likely trajectory direction, lineage ambiguity, growth, or insufficient information in snapshot expression than the absence of a pseudotime coordinate.

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